

Discussion: Cross-Domain Analysis of Cognitive Integrity

Our cross-domain analysis of ten critical sectors reveals that Goal Hijacking is not merely a linguistic exploit but a structural corruption of the OODA Loop [?]. In every case—from drone swarms operating at millisecond time scales to diplomatic agents spanning months of deliberation—the attack vector was a transient signal that hijacked the agent's **Orientation** phase, rewriting its Functional Requirements in real-time. This section synthesizes the cross-domain findings, identifies universal attack patterns, evaluates CIF mechanism coverage, and acknowledges limitations.

4.1 Cross-Domain Attack Pattern Taxonomy

Three universal attack patterns emerge across the ten domains. Each pattern corresponds to a distinct manipulation of the Axiomatic Design Matrix [?]:

Pattern 1: FR Polarity Inversion. The adversary flips the sign of a Functional Requirement, transforming a minimization objective into a maximization objective (or vice versa). The diagonal element A_{ii} effectively changes sign. This is the most common pattern, appearing in five domains.